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Safety data sheet

BASF Safety data sheet according to UN GHS 4th rev.
Date / Revised: 06.01.2017 Version: 2.1
Product: **Lucantin® CX forte**
(ID no. 30041155/SDS_GEN_00/EN)
Date of print 05.07.2025

1. Identification

Product identifier

Lucantin® CX forte

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: feed additive(s)

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Nutrition and Health

Telephone: +49 621 60-48434
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2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Skin Sens. 1

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Warning

Hazard Statement:

H317	May cause an allergic skin reaction.

Precautionary Statements (Prevention):

P280	Wear protective gloves.
P260	Do not breathe dust/gas/mist/vapours.
P272	Contaminated work clothing should not be allowed out of the workplace.

Precautionary Statements (Response):

P311	Call a POISON CENTER or doctor/physician.
P303 + P352	IF ON SKIN (or hair): Wash with plenty of soap and water.
P362 + P364	Take off contaminated clothing and wash it before reuse.

Precautionary Statements (Disposal):

P501	Dispose of contents/container to hazardous or special waste collection point.

Other hazards

According to UN GHS criteria

The product is under certain conditions capable of dust explosion. The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

Preparation based on: (E)-5,9,14,18-Tetramethyl-20-(2,6,6-trimethylcyclohexenyl)-3,5,7,9,11,13,15,17,19-icosanonaen-2-one

in a matrix of: carbohydrates, Gelatins

stabilized with: ethoxyquin(ISO); 6-ethoxy-1,2-dihydro-2,2,4-trimethylquinoline

Hazardous ingredients (GHS)
According to UN GHS criteria

ethoxyquin(ISO); 6-ethoxy-1,2-dihydro-2,2,4-trimethylquinoline

Content (W/W): >= 3 % - < 5 %	Acute Tox. 4 (oral)
CAS Number: 91-53-2	H302
INDEX-Number: 613-014-00-2	

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

Immediately remove contaminated clothing. If adverse health effects develop seek medical attention.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: The most important known symptoms and effects are described in the labelling (see section 2) and/or in section 11., Further important symptoms and effects are so far not known.

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
water spray, carbon dioxide, foam, dry powder

Unsuitable extinguishing media for safety reasons:
water jet

Additional information:
Avoid whirling up the material/product because of the danger of dust explosion.

Special hazards arising from the substance or mixture
carbon oxides
Evolution of fumes/fog.

Advice for fire-fighters
Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.
Cool endangered containers with water-spray.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures
Do not breathe dust. Avoid dust formation. Ensure adequate ventilation. Use personal protective clothing. Information regarding personal protective measures see, section 8.

Environmental precautions
Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up
For small amounts: Pick up with suitable appliance and dispose of.
For large amounts: Contain with dust binding material and dispose of.
Dispose of absorbed material in accordance with regulations. Avoid raising dust. Cleaning operations should be carried out only while wearing breathing apparatus.

Reference to other sections
Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

7. Handling and Storage

Precautions for safe handling
Avoid dust formation. Provide exhaust ventilation if dust is formed.

Protection against fire and explosion:

Avoid dust formation. The product is capable of dust explosion. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy. Use explosion-proof apparatus and fittings.

Conditions for safe storage, including any incompatibilities

Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

57-50-1: Sucrose
91-53-2: ethoxyquin(ISO); 6-ethoxy-1,2-dihydro-2,2,4-trimethylquinoline
8001-21-6: Sunflower oil
9000-70-8: Gelatins

Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if breathable aerosols/dust are formed. Particle filter with low efficiency for solid particles (e.g. EN 143 or 149, Type P1or FFP1)

Hand protection:
Suitable chemical resistant safety gloves (EN 374) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.
Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Avoid contact with the substance. Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is required additionally to the stated personal protection equipment. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Store work clothing separately.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	powder
Colour:	orange to red
Odour:	faint specific odour
Odour threshold:	not determined
pH value:	approx. 6 (50 g/l, 20 °C)
Melting point:	not determined
Boiling point:	not relevant
Flash point:	not applicable, the product is a solid
Evaporation rate:	The product is a non-volatile solid.
Flammability:	not flammable
Lower explosion limit:	For solids not relevant for classification and labelling.
Upper explosion limit:	For solids not relevant for classification and labelling.
Vapour pressure:	negligible
Relative vapour density (air):	The product is a non-volatile solid.
Solubility in water:	dispersible (> 35 °C)
Partitioning coefficient n-octanol/water (log Kow):	not applicable for mixtures

Information on: ethoxyquin(ISO); 6-ethoxy-1,2-dihydro-2,2,4-trimethylquinoline
Partitioning coefficient n-octanol/water (log Kow): 3,87
(25 °C)

Thermal decomposition:	115 °C, 220 kJ/kg (DSC (DIN 51007)) self-accelerating reaction
Viscosity, dynamic:	not applicable, the product is a solid

Explosion hazard:	Product is not explosive, however a dust explosion could result from an air / dust mixture.	
Fire promoting properties:	Based on its structural properties the product is not classified as oxidizing.	

Other information

Burning rate:	0,03 mm/s The material doesn't meet the criteria specified in paragraph 33.2.1.4.4 of UN manual of tests and criteria.	(Directive 92/69/EEC, A.10)
Self heating ability:	It is not a substance capable of spontaneous heating.	(VDI 2263, sheet 1, 1.4.1)
	It is not a substance capable of spontaneous heating.	(VDI 2263, sheet 1, 1.4.2)
Minimum ignition energy:	> 1 J The product is capable of dust explosion.	(DIN EN 13821)
Bulk density:	approx. 650 kg/m3	

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: Corrosive effects to metal are not anticipated.

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Peroxides: The product does not contain peroxides.

Possibility of hazardous reactions
Dust explosion hazard.

Conditions to avoid
Avoid all sources of ignition: heat, sparks, open flame. See MSDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
None known during use and storage if used according to instructions.

Hazardous decomposition products

:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion.

Experimental/calculated data:
LD50 rat (oral): > 6.400 mg/kg (BASF-Test)

Information on: (E)-5,9,14,18-Tetramethyl-20-(2,6,6-trimethylcyclohexenyl)-3,5,7,9,11,13,15,17,19-icosanonaen-2-one
Assessment of acute toxicity:
Virtually nontoxic after a single ingestion.

Information on: ethoxyquin(ISO); 6-ethoxy-1,2-dihydro-2,2,4-trimethylquinoline
Assessment of acute toxicity:
Of moderate toxicity after single ingestion. Virtually nontoxic after a single skin contact.

Irritation

Assessment of irritating effects:
Not irritating to eyes and skin.

Information on: (E)-5,9,14,18-Tetramethyl-20-(2,6,6-trimethylcyclohexenyl)-3,5,7,9,11,13,15,17,19-icosanonaen-2-one
Assessment of irritating effects:
Not irritating to the skin. Not irritating to the eyes.

Information on: ethoxyquin(ISO); 6-ethoxy-1,2-dihydro-2,2,4-trimethylquinoline
Assessment of irritating effects:
May cause slight irritation to the skin. May cause slight irritation to the eyes.

Respiratory/Skin sensitization

Assessment of sensitization:
Caused skin sensitization in animal studies.

Experimental/calculated data:

Mouse Local Lymph Node Assay (LLNA) mouse: skin sensitizing (OECD Guideline 429)
Tested as a preparation.

Germ cell mutagenicity

Assessment of mutagenicity:
No data available.

Information on: (E)-5,9,14,18-Tetramethyl-20-(2,6,6-trimethylcyclohexenyl)-3,5,7,9,11,13,15,17,19-icosanonaen-2-one
Assessment of mutagenicity:
No mutagenic effect was found in various tests with bacteria and mammals.

Carcinogenicity

Assessment of carcinogenicity:
No data available.

Information on: (E)-5,9,14,18-Tetramethyl-20-(2,6,6-trimethylcyclohexenyl)-3,5,7,9,11,13,15,17,19-icosanonaen-2-one
Assessment of carcinogenicity:
In long-term studies in rats in which the substance was given by feed, a carcinogenic effect was not observed.

Reproductive toxicity

Assessment of reproduction toxicity:
No data available.

Information on: (E)-5,9,14,18-Tetramethyl-20-(2,6,6-trimethylcyclohexenyl)-3,5,7,9,11,13,15,17,19-icosanonaen-2-one
Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
No data available.

Information on: (E)-5,9,14,18-Tetramethyl-20-(2,6,6-trimethylcyclohexenyl)-3,5,7,9,11,13,15,17,19-icosanonaen-2-one
Assessment of teratogenicity:
No indications of a developmental toxic / teratogenic effect were seen in animal studies.

Specific target organ toxicity (single exposure)

Remarks: not applicable

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on available Data, the classification criteria are not met.

Aspiration hazard

not applicable

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
No data available.

Information on: (E)-5,9,14,18-Tetramethyl-20-(2,6,6-trimethylcyclohexenyl)-3,5,7,9,11,13,15,17,19-icosanonaen-2-one
Toxicity to fish:
LC50 (96 h) > 100 mg/l, Brachydanio rerio (OECD 203; ISO 7346; 92/69/EEC, C.1, static)
The product has low solubility in the test medium. An eluate has been tested. The details of the toxic effect relate to the nominal concentration. No toxic effects occur within the range of solubility. Limit concentration test only (LIMIT test).

Information on: ethoxyquin(ISO); 6-ethoxy-1,2-dihydro-2,2,4-trimethylquinoline
Toxicity to fish:
LC50 (96 h) 18 mg/l, Oncorhynchus mykiss (OPP 72-1 (EPA-Guideline), Flow through.)
The details of the toxic effect relate to the nominal concentration.

Persistence and degradability

Assessment biodegradation and elimination (H2O):
No data available.

Information on: ethoxyquin(ISO); 6-ethoxy-1,2-dihydro-2,2,4-trimethylquinoline
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria). Poorly biodegradable.

Information on: (E)-5,9,14,18-Tetramethyl-20-(2,6,6-trimethylcyclohexenyl)-3,5,7,9,11,13,15,17,19-icosanonaen-2-one
Assessment biodegradation and elimination (H2O):
Poorly biodegradable. Not readily biodegradable (by OECD criteria).

Bioaccumulative potential

Assessment bioaccumulation potential:

The product has not been tested.

Information on: ethoxyquin(ISO); 6-ethoxy-1,2-dihydro-2,2,4-trimethylquinoline
Assessment bioaccumulation potential:
No significant accumulation in organisms is expected as a result of the distribution coefficient of n-octanol/water (log Pow).

Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: Adsorption to solid soil phase is expected.

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

13. Disposal Considerations

Waste treatment methods

Observe national and local legal requirements.

14. Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for	None known

user

Inland waterway transport

ADN

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel

Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

UN number

See corresponding entries for “UN number” for the respective regulations in the tables above.

UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

Transport in bulk according to Annex II of MARPOL and the IBC Code

Regulation:	Not evaluated
Shipment approved:	Not evaluated
Pollution name:	Not evaluated
Pollution category:	Not evaluated
Ship Type:	Not evaluated

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

Any other intended applications should be discussed with the manufacturer. Corresponding occupational protection measurements must be followed.

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Skin Sens.	Skin sensitization
Acute Tox.	Acute toxicity
H302	Harmful if swallowed.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.